

Lecture 19: Managing Missing Data

Heather Mattie, PhD

November 4, 2025

Recipe of the Day!

Turkish Çiğ Köfte



Announcements

- No lab this week
- Midterm due **Sunday, November 9**
 - No late days may be used
- No class November 11, 25, 27



Homework #3 clarifications

- The measles vaccine was introduced in the US in 1963, but wasn't introduced in California until 1966. Plotting either or both lines is fine.
- You can square a sequence of numbers for the tick marks in question 4.

```
data %>%  
  filter(state == "California", !is.na(rate)) %>%  
  ggplot(aes(x = year, y = rate)) +  
  geom_line(alpha = 0.5) +  
  geom_point() +  
  scale_y_sqrt("Cases per 100,000", breaks = seq(0, 100, 2)^2) +  
  xlab("Year") +  
  ggtitle("Measles Cases in California") +  
  geom_vline(xintercept = 1963, col = "blue") +  
  geom_text(aes(x = 1980, y = 1500, label = "Measles Vaccine Introduced in 1963"), color = "blue")
```

Homework #3 clarifications

- Calculating the US rate over time – there are multiple ways to calculate it

```
avg <- data %>%  
  filter(weeks_reporting > 0) %>%  
  group_by(year) %>%  
  summarize(us_rate = sum(count / weeks_reporting) * 52 / sum(population / 100000))
```


Coding question of the day

Below, a link to a dataset available on GitHub is provided. The data comes from the [Johns Hopkins University COVID-19 data repository](#). This particular dataset contains reported COVID-19 case counts from countries around the globe from January 22, 2020 to March 23, 2020.

Read in the data using the code provided. This dataset is currently in wide format. Convert the data frame to long format, with a column named `date` and a column named `cases`. Then, make the date column values into date objects.

Bonus challenge: rename the `Country/Region` column `country`. Create a new data frame with country name, date, and the total number of cases in the country on each day.

```
library(readr)
library(dplyr)
library(tidyr)
library(lubridate)

data <- read_csv("https://raw.githubusercontent.com/CSSEGISandData/COVID-19/refs/heads/master/archived_data/archived_time_series/time_series_19-covid-Recovered_archived_0325.csv")
head(data)
```

Managing missing data

Heather Mattie, PhD

Managing Missing Data Roadmap

1. Types of missingness
2. Simple imputation methods
3. Advanced imputation methods
4. Survey skip patterns

Managing missing data

Types of missingness

Heather Mattie, PhD

Missingness mechanisms

There are four types of missingness

1. Missing completely at random (MCAR)
2. Missing at random (MAR)
3. Not missing at random (NMAR)
4. Missingness by design

Understanding patterns of missingness is a crucial first step in handling missing values

		?	
?			
?	?		
			?
		?	

MCAR

Missing completely at random (**MCAR**) means that there is no relationship between the probability of data being missing and any of the other variables in the data

- The probability of being missing is the same for all cases
- Rare in public health research

		?	
?			
?	?		
			?
		?	

MCAR

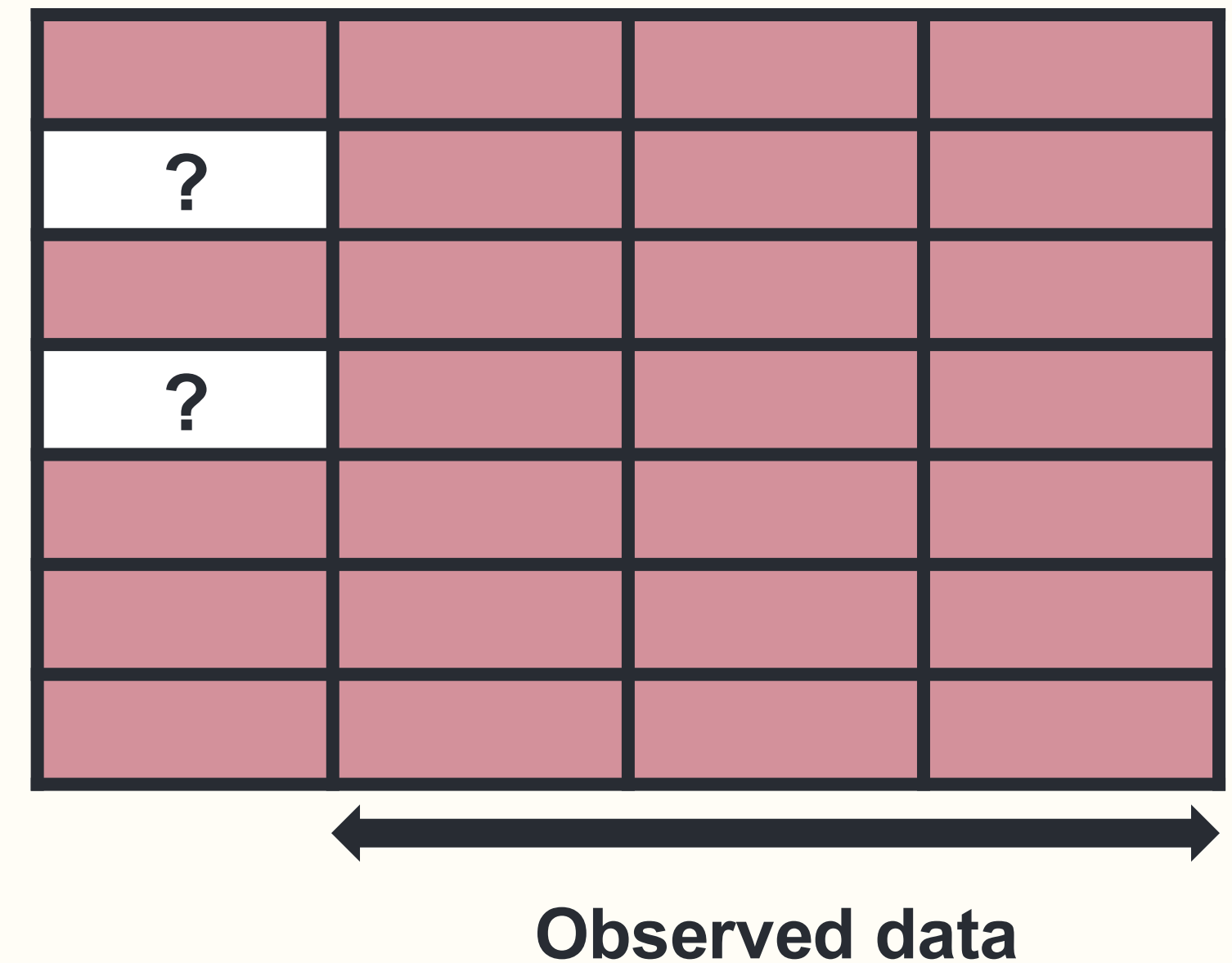
- **Example:** a patient's blood pressure reading missing due to device malfunction, regardless of other patient information
- **Solution:** simple imputation methods (mean, median, mode, LOCF, BOCF)

		?	
?			
?	?		
			?
		?	

MAR

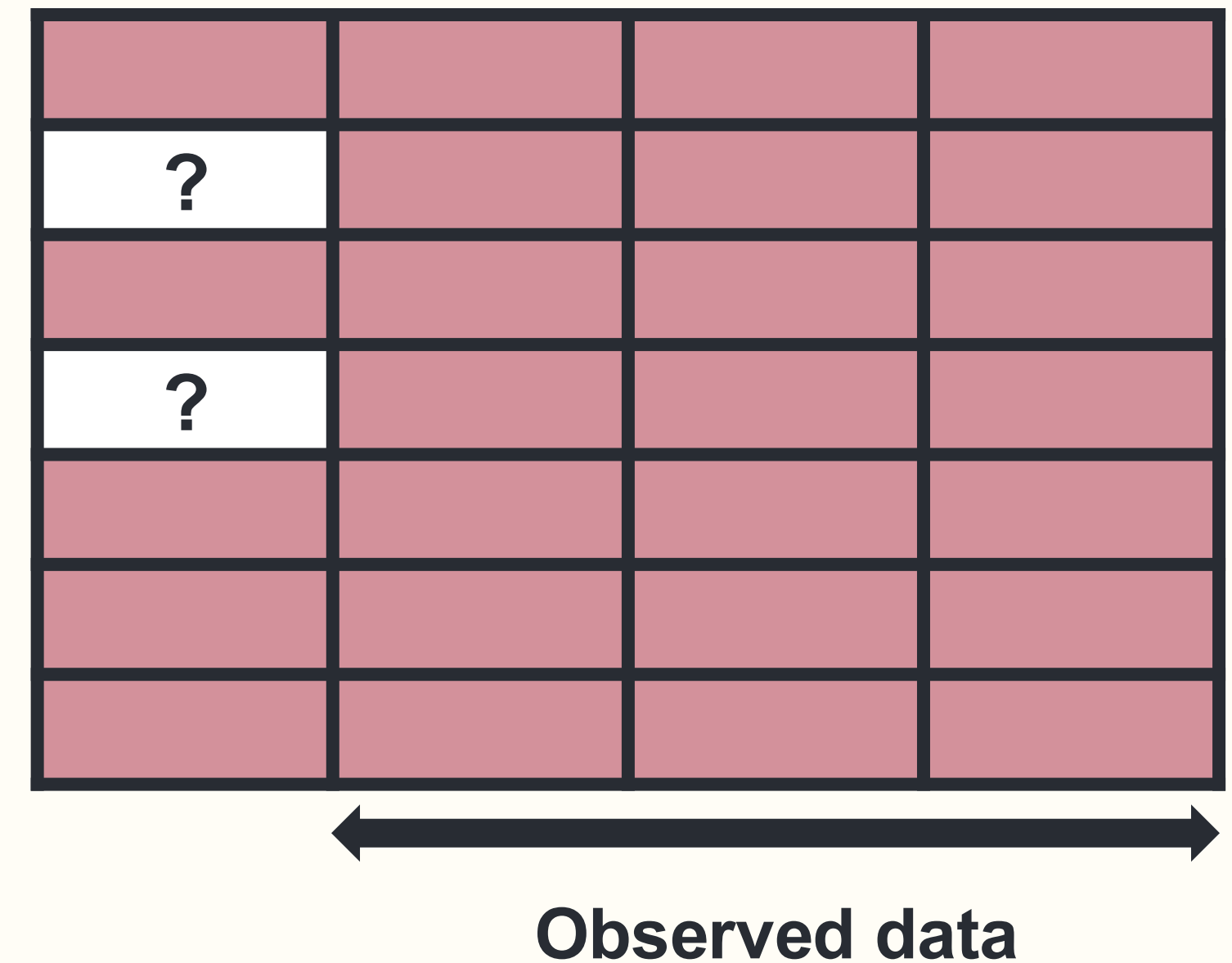
Missing at random (**MAR**) means that the data is missing in a systematic, predictable way based on the other information (variables) in the dataset

- Missingness is not related to the value that is missing itself
- Common in public health research



MAR

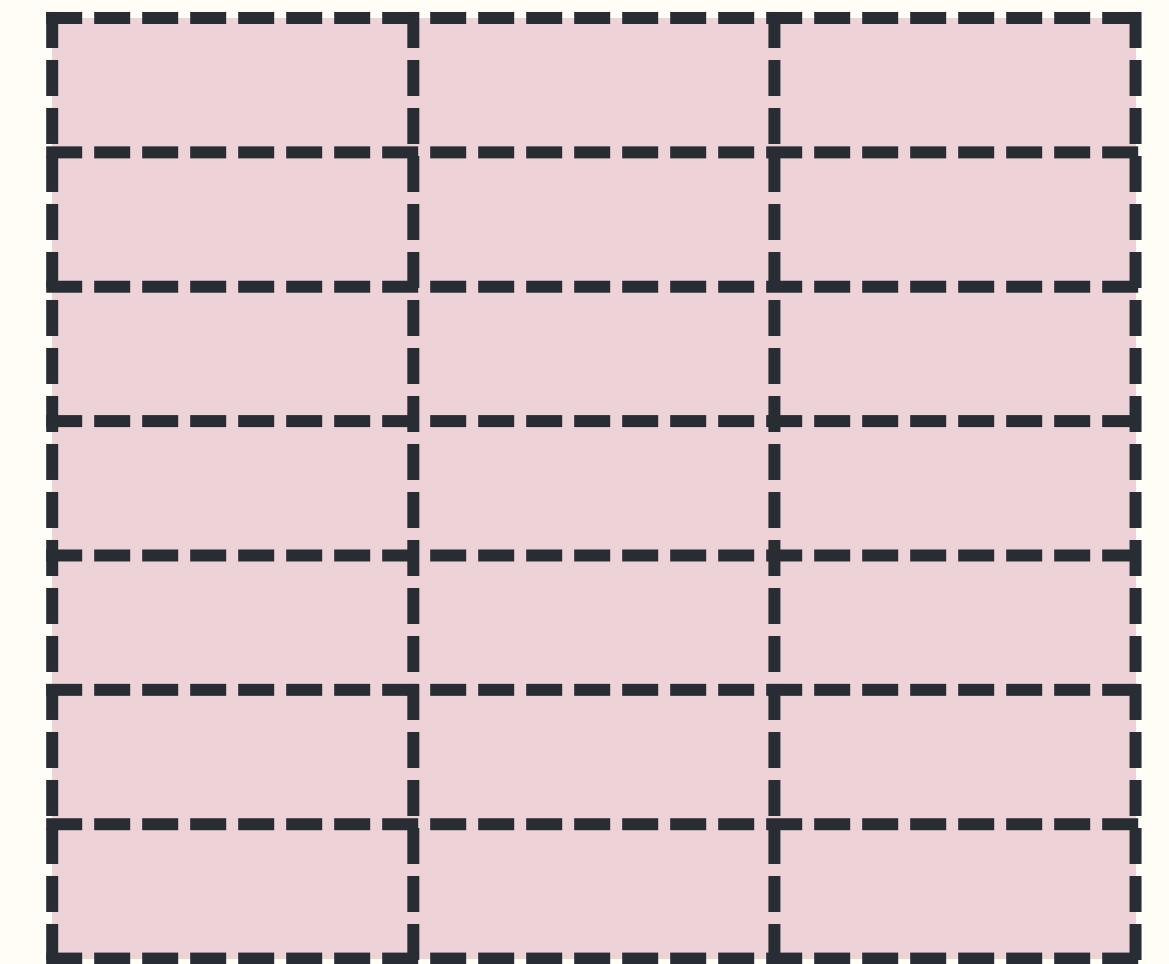
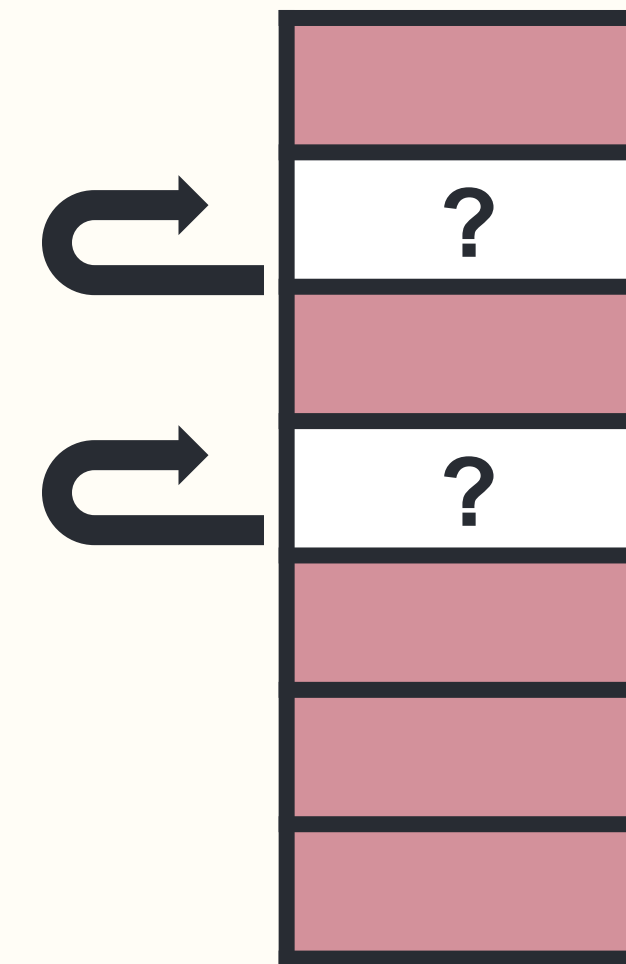
- Example: the presence of a fever missing because all patients with chills and aches were assumed to have a fever and their temperature was not taken
- Solution: multiple imputation



NMAR

Not missing at random (**NMAR**) means that the data is NOT missing in a systematic, predictable way based on the other information (variables) in the dataset, but also isn't missing randomly

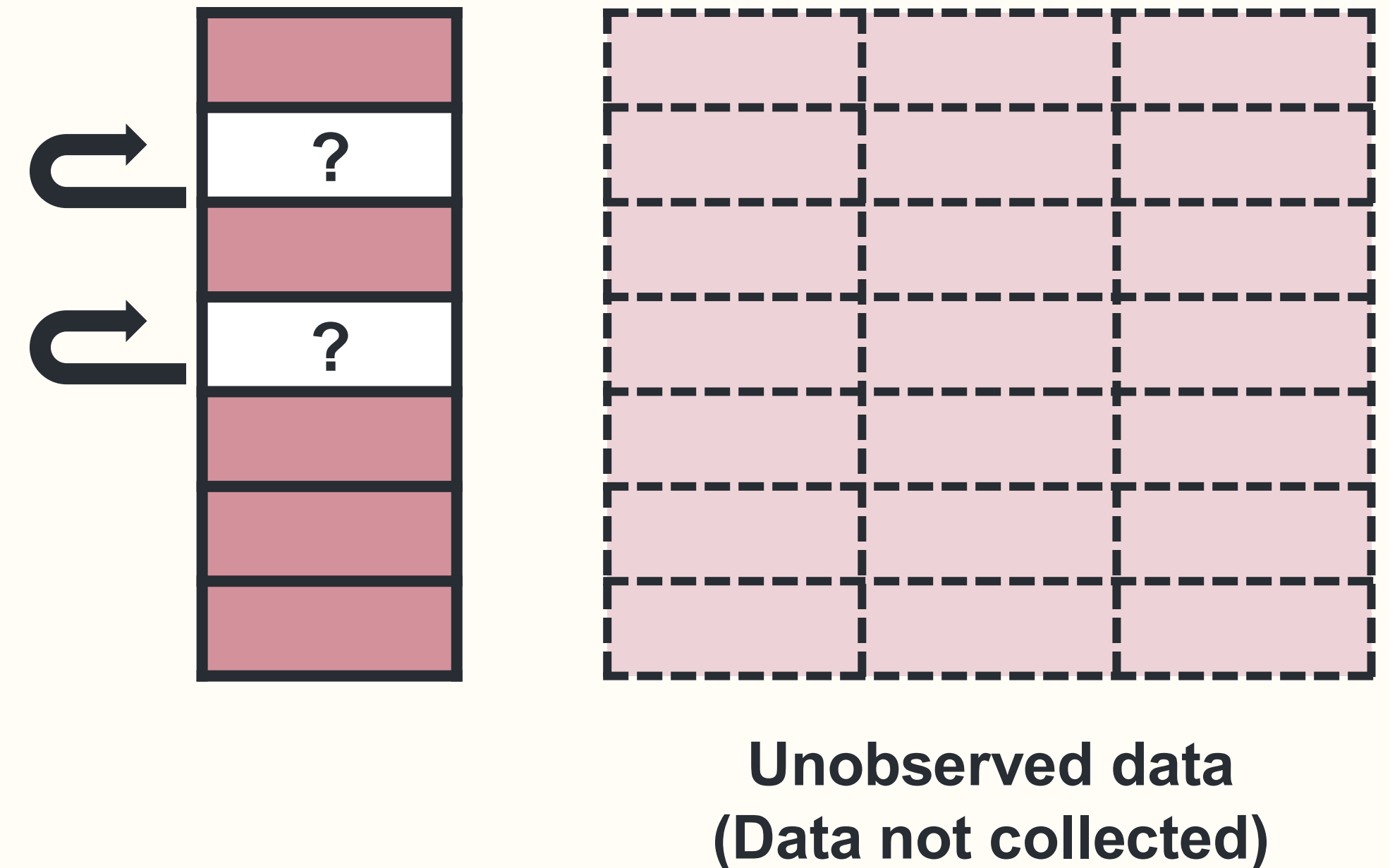
- Missingness is attributed to the missing value itself or the variables not collected
- There is a pattern to the missingness, but it isn't due to any observed data



Unobserved data
(Data not collected)

NMAR

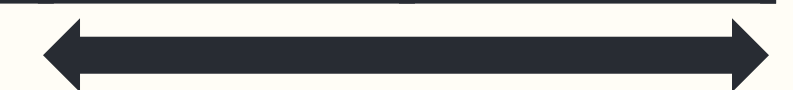
- **Example:** the age of very elderly patients missing because they don't know it or refuse to say how old they are
- **Solution:** Try to collect more data or information about why data is missing; imputation is usually not recommended



Missingness by design

Missingness by design refers to intentionally leaving out data according to a predefined plan

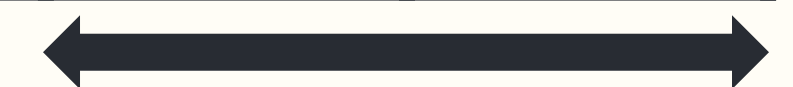
- Common in surveys
- Often used to reduce participant burden or data collection costs
- Missing values can typically be imputed in a straightforward way



These questions
skipped based on
response to
previous question

Missingness by design

- **Example:** respondents of a survey skipping questions about smoking habits if they are non-smokers
- **Solution:** skip pattern imputation



These questions
skipped based on
response to
previous question

Summary

- There are four types of missingness: MCAR, MAR, NMAR, and missingness by design
- Ignoring the missingness mechanism can lead to biased or misleading results
- Each type of missingness should be handled in a different way

Managing missing data

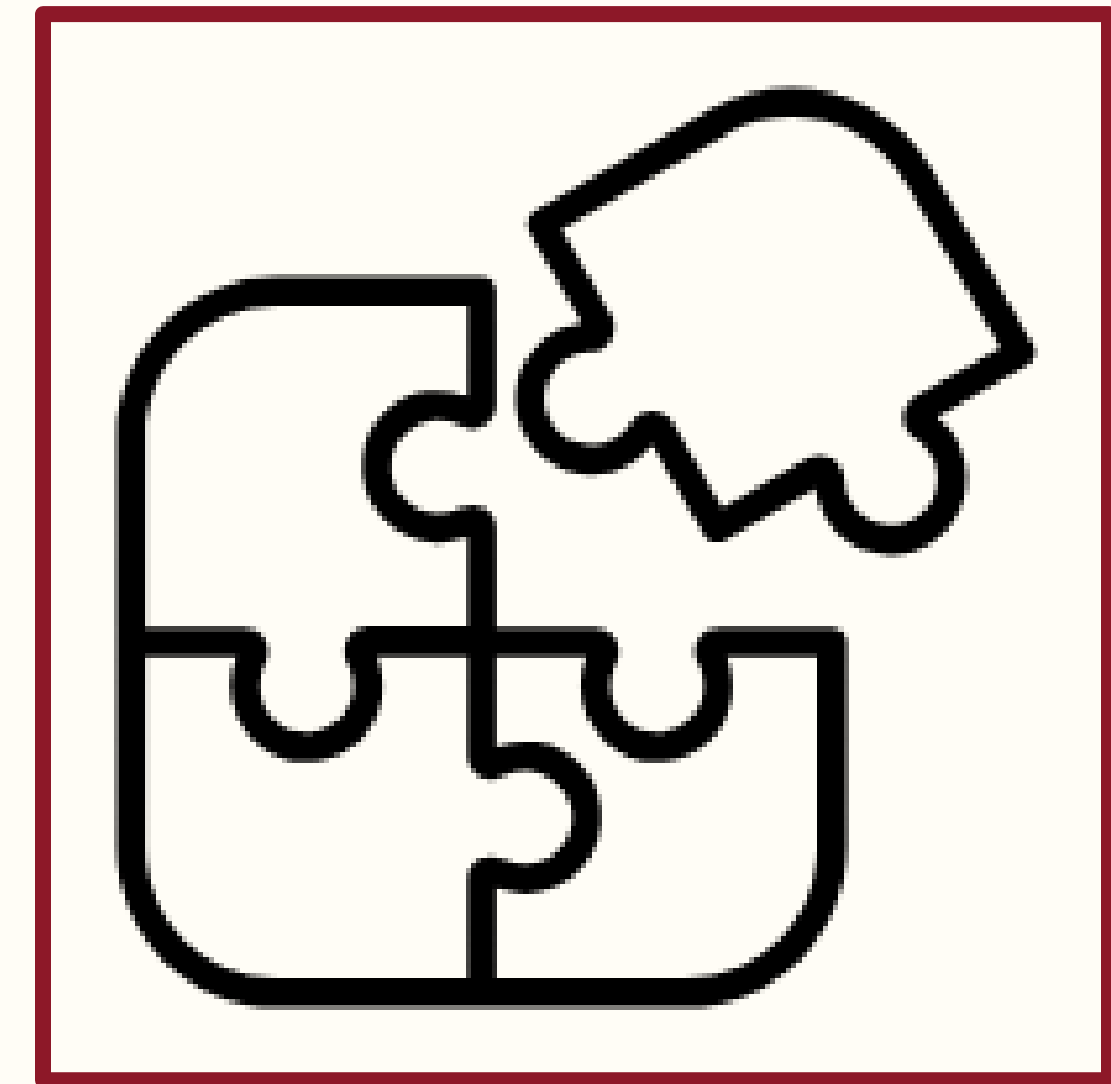
Simple imputation methods

Heather Mattie, PhD

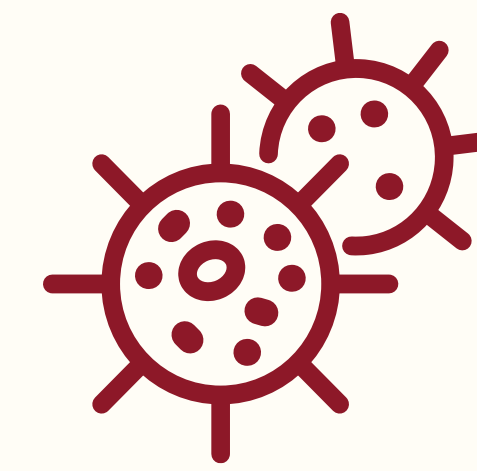
Simple imputation methods

When it can be assumed the type of missingness is MCAR, **simple imputation** methods can be used

- Methods we will cover include mean, median, mode, LOCF, and BOCF
- MCAR is a strong assumption and simple imputation methods should be used with caution



Simulated Ebola outbreak dataset



Simulated data mimicking the West African Ebola outbreak of 2014-2015

- **fever**: indicator of fever
- **temp**: temperature in degrees Celsius
- **outcome**: indicator of recovery or death

```
ebola_subset <- ebola %>%  
  select(temp, fever, outcome)  
  
head(ebola_subset)
```

```
##    temp fever outcome  
## 1 36.8    no    <NA>  
## 2 36.9  <NA> Recover  
## 3 36.9  <NA> Recover  
## 4 36.8    no    <NA>  
## 5 36.9    no Recover  
## 6 37.6    no Recover
```

```
colSums(is.na(ebola_subset))
```

```
##      temp      fever outcome  
##      149       249     1323
```

Mean imputation

Mean imputation replaces missing values in a column with the mean of the non-missing values in the same column

- For numeric values only
- Can use the **mean** function

```
ebola <- ebola %>%  
  mutate(temp_replace_na_with_mean =  
    replace_na(temp, mean(temp, na.rm = TRUE)))  
  
mean(ebola$temp, na.rm = TRUE)
```

```
## [1] 38.55829
```

```
mean(ebola$temp_replace_na_with_mean)
```

```
## [1] 38.55829
```


Median imputation

Median imputation replaces missing values in a column with the median of the non-missing values in the same column

- For numeric or categorical values
- Can use the **median** function

```
ebola <- ebola %>%  
  mutate(temp_replace_na_with_median =  
    replace_na(temp, median(temp, na.rm = TRUE)))  
  
median(ebola$temp, na.rm = TRUE)
```

```
## [1] 38.8
```

```
median(ebola$temp_replace_na_with_median)
```

```
## [1] 38.8
```


Mode imputation

Mode imputation replaces missing values in a column with the mode of the non-missing values in the same column

- Primarily used for categorical variables
- Can use the **impute_mode** function from the **missMethods** package

```
library(missMethods)
```

```
table(ebola$fever, useNA = "ifany")
```

```
##
```

```
##   no  yes <NA>
```

```
## 1090 4549  249
```

```
ebola$fever <- impute_mode(as.data.frame(ebola$fever))
```

```
table(ebola$fever, useNA = "ifany")
```

```
## ebola$fever
```

```
##   no  yes
```

```
## 1090 4798
```

LOCF and BOCF

Last observation carried forward (LOCF) and baseline observation carried forward (BOCF) are imputation methods for handling missing data in longitudinal studies

- **LOCF** replaces a missing value with the last recorded (observed) value for that variable for a particular person
- **BOCF** replaces all missing values of a participant's variable with the baseline (first) measurement

LOCF and BOCF

Last observation carried forward (LOCF) and baseline observation carried forward (BOCF) are imputation methods for handling missing data in longitudinal studies

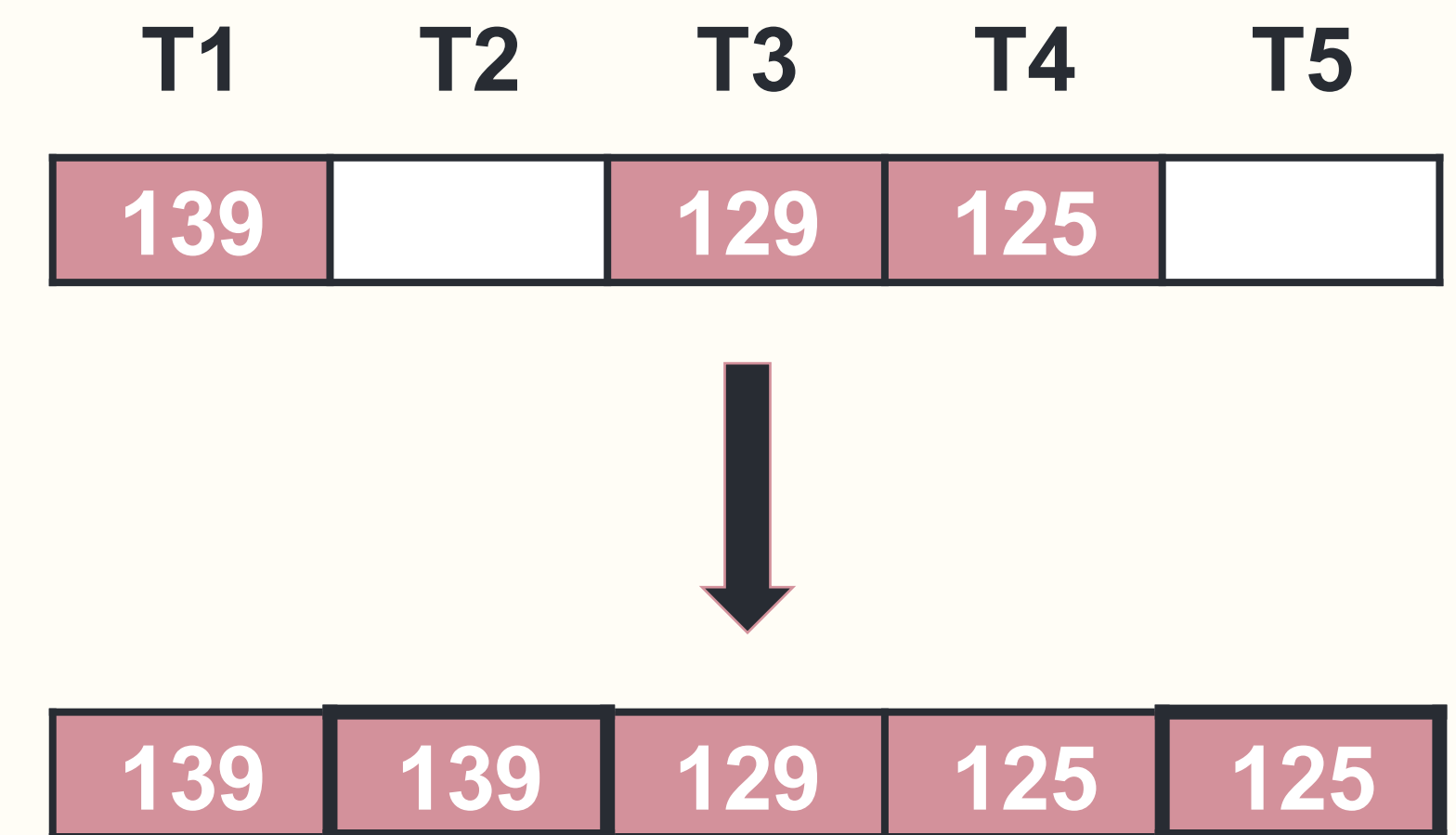
- **LOCF** replaces a missing value with the last recorded (observed) value for that variable for a particular person
- BOCF replaces all missing values of a participant's variable with the baseline (first) measurement

T1	T2	T3	T4	T5
139		129	125	

LOCF and BOCF

Last observation carried forward (LOCF) and baseline observation carried forward (BOCF) are imputation methods for handling missing data in longitudinal studies

- **LOCF** replaces a missing value with the last recorded (observed) value for that variable for a particular person
- BOCF replaces all missing values of a participant's variable with the baseline (first) measurement



LOCF and BOCF

Last observation carried forward (LOCF) and baseline observation carried forward (BOCF) are imputation methods for handling missing data in longitudinal studies

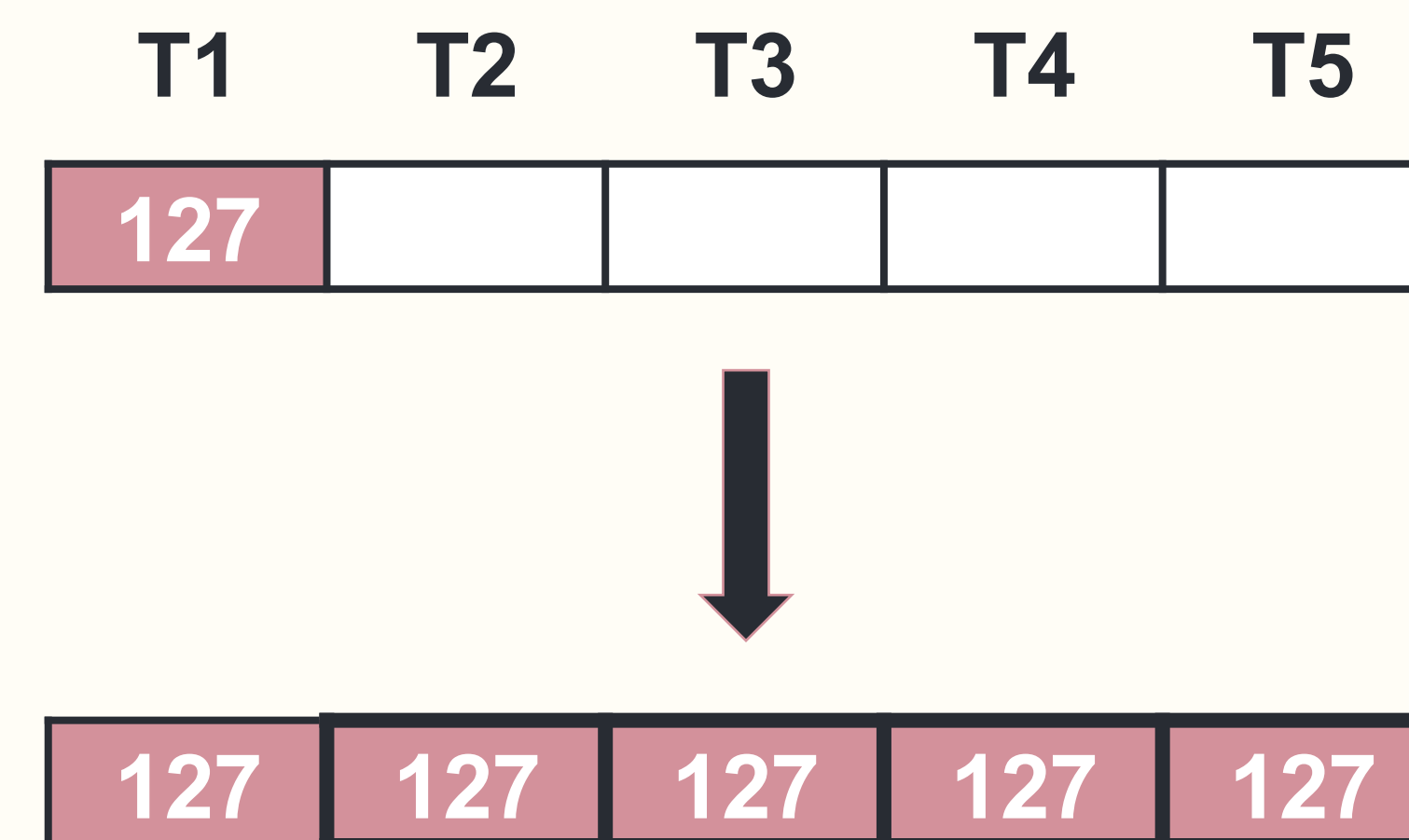
- LOCF replaces a missing value with the last recorded (observed) value for that variable for a particular person
- **BOCF** replaces all missing values of a participant's variable with the baseline (first) measurement

T1	T2	T3	T4	T5
127				

LOCF and BOCF

Last observation carried forward (LOCF) and baseline observation carried forward (BOCF) are imputation methods for handling missing data in longitudinal studies

- LOCF replaces a missing value with the last recorded (observed) value for that variable for a particular person
- **BOCF** replaces all missing values of a participant's variable with the baseline (first) measurement



LOCF and BOCF

data

##	patID	visit	sbp
## 1	1	1	139
## 2	1	2	NA
## 3	1	3	129
## 4	1	4	125
## 5	1	5	NA
## 6	2	1	127
## 7	2	2	NA
## 8	2	3	NA
## 9	2	4	NA
## 10	2	5	NA



```
data <- data %>% fill(sbp)
print(data)
```

##	patID	visit	sbp
## 1	1	1	139
## 2	1	2	139
## 3	1	3	129
## 4	1	4	125
## 5	1	5	125
## 6	2	1	127
## 7	2	2	127
## 8	2	3	127
## 9	2	4	127
## 10	2	5	127

Summary

- Simple imputation methods can be used when the data missingness mechanism is MCAR
- Mean imputation and median imputation are common methods for numerical data
- Mode imputation is common for categorical data
- LOCF and BOCF are used for longitudinal data

Managing missing data

Advanced imputation methods

Heather Mattie, PhD

Advanced imputation methods

When it can be assumed the type of missingness is MAR, **advanced imputation** methods are needed

- Methods we will cover include regression imputation and multiple imputation
- These methods take the missingness mechanism into account and reduce some of the bias that imputation can induce

Regression imputation

Regression imputation uses a statistical model to predict what a missing value is likely to be and replaces it with the predicted value

- Other variables in the dataset are used to build the model
- “Regression” means any type of regression model (e.g., linear, logistic, etc.)
- Should be used with caution because variability of imputed data may be underestimated, even if the model is very good

?			
?			

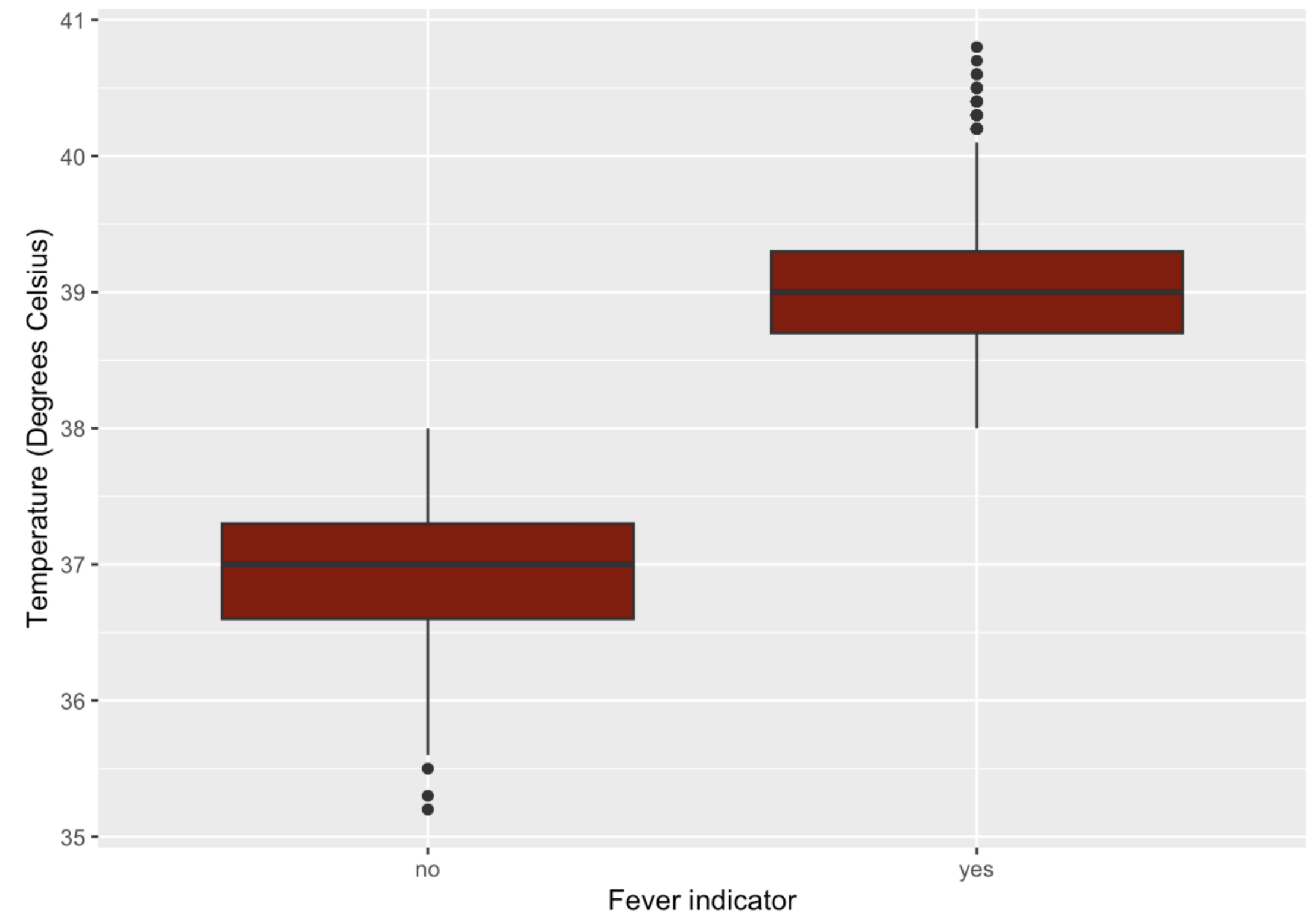
These variables are put into a model to predict missing values in the first column

Regression imputation example

Using the Ebola dataset, fever and age will be used in a linear regression model to predict missing values for temperature

- Linear regression is used since temperature is a continuous variable
- Plausible that temperature was not taken if patient presented with chills and aches

```
ebola %>% ggplot(aes(fever, temp)) +  
  geom_boxplot(fill = "darkred") +  
  xlab("Fever indicator") +  
  ylab("Temperature (Degrees Celsius)")
```

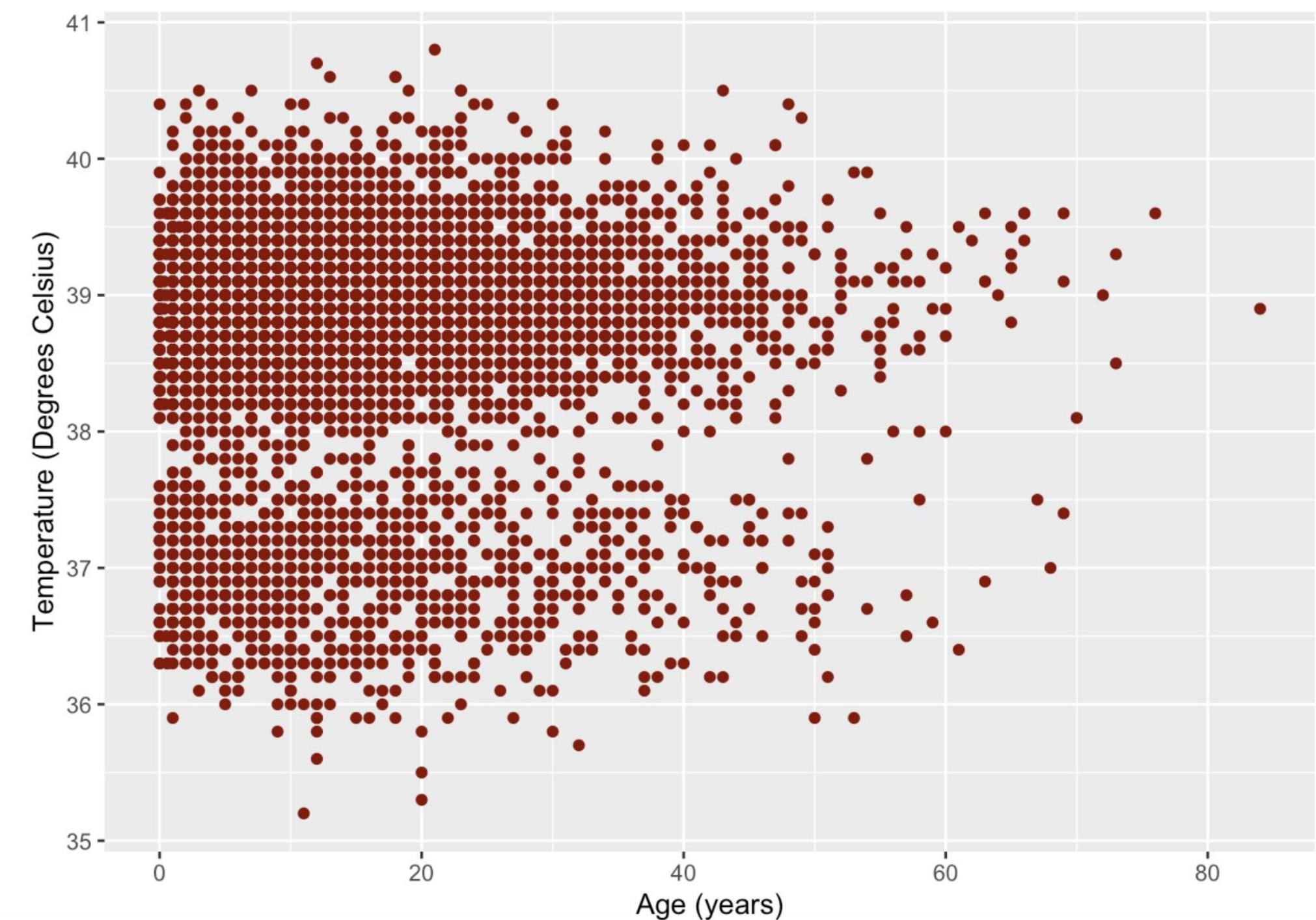


Regression imputation example

Using the Ebola dataset, age and fever will be used in a linear regression model to predict missing values for temperature

- Linear regression is used since temperature is a continuous variable
- Plausible that temperature was not taken if patient presented with chills and aches

```
ebola <- ebola %>%  
  filter(!is.na(fever) & !is.na(age_years))  
  
ebola %>% ggplot(aes(age_years, temp)) +  
  geom_point(color = "darkred") +  
  xlab("Age (years)") +  
  ylab("Temperature (Degrees Celsius)")
```



Regression imputation example

Using the Ebola dataset, age and fever will be used in a linear regression model to predict missing values for temperature

- Linear regression is used since temperature is a continuous variable
- Plausible that temperature was not taken if patient presented with chills and aches

```
reg_model <- lm(temp ~ fever + age_years, data = ebola)

newdata <- ebola %>% filter(is.na(temp))

temps_predictions <- predict(reg_model, newdata = newdata)
head(temps_predictions)
```

```
##           1           2           3           4           5           6
## 36.97721 36.97175 36.96728 36.97970 36.97622 36.97622
```

Multiple imputation

Why stop with one imputed value per missing value?

Multiple imputation creates multiple datasets with the missing values imputed to plausible data values

- Keep observed values the same in each dataset
- Reduces bias and results in more accurate standard error estimates

		?	
?			
?	?		
			?

		?	
?			
?	?		
			?

		?	
?			
?	?		
			?

		?	
?			
?	?		
			?

Multiple imputation

Steps:

1. Initialization:

- Missing values are replaced with plausible values (usually the mean from the observed data)

		?	
?			
?	?		
			?

		?	
?			
?	?		
			?

		?	
?			
?	?		
			?

		?	
?			
?	?		
			?

Multiple imputation

2. Iterative imputation:

- Cycle through each variable with missing values
- Fit a predictive model using the other variables as predictors
- Use model to predict missing values for that variable
- The imputed values are used in subsequent iterations for other variables

The figure displays four 5x4 grids, each representing a data matrix with missing values (indicated by question marks). The grids illustrate the iterative imputation process, showing how missing values are predicted and then used in subsequent iterations for other variables.

Grid 1 (Top Left):

		?	
?			
?	?		
			?

Grid 2 (Top Right):

		?	
?			
?	?		
			?

Grid 3 (Bottom Left):

		?	
?			
?	?		
			?

Grid 4 (Bottom Right):

		?	
?			
?	?		
			?

Multiple imputation

3. Convergence:

- Iterative process continues until imputed values stabilize

4. Multiple datasets:

- Repeat the process multiple times to achieve multiple datasets

5. Analysis and pooling:

- Analyze each imputed dataset separately, and then pool the results

		?	
?			
?	?		
			?

		?	
?			
?	?		
			?

		?	
?			
?	?		
			?

		?	
?			
?	?		
			?

mice

The **mice** function can be used to perform multiple imputation

- Stands for “multiple imputation by chained equations”
- Can impute numerical data, binary data, unordered categorical data, and ordered categorical data



mice

```
library(mice)

model_dataset <- ebola %>%
  select(temp, fever, age_years)

md.pattern(model_dataset)
```

	fever	age_years	temp	
5404				0
149				1
	0	0	149	149

mice

```
multiple_imputation = mice(  
  model_dataset,  
  seed = 1,  
  m = 10,  
  print = FALSE)
```

```
model_fit <- with(multiple_imputation, lm(temp ~ age_years + fever))  
  
summary(mice::pool(model_fit))
```

```
##           term      estimate  std.error  statistic      df      p.value  
## 1 (Intercept) 3.702800e+01 0.0258516265 1432.3277729 31.99099 1.797488e-78  
## 2   age_years 2.344859e-04 0.0006359435    0.3687213 110.48345 7.130411e-01  
## 3   feveryes 1.978807e+00 0.0191893059  103.1203259 227.16366 9.200270e-193
```

mice

```
model_fit <- with(multiple_imputation, lm(temp ~ age_years + fever))

summary(mice::pool(model_fit))
```

```
##           term      estimate  std.error  statistic      df      p.value
## 1 (Intercept) 3.702800e+01 0.0258516265 1432.3277729  31.99099 1.797488e-78
## 2   age_years 2.344859e-04 0.0006359435    0.3687213 110.48345 7.130411e-01
## 3   feveryes 1.978807e+00 0.0191893059  103.1203259 227.16366 9.200270e-193
```

```
complete_data <- model_dataset[complete.cases(model_dataset),]

model_fit <- with(complete_data, lm(temp ~ age_years + fever))

tidy(summary(model_fit))
```

```
## # A tibble: 3 × 5
##   term      estimate std.error statistic p.value
##   <chr>      <dbl>    <dbl>    <dbl>    <dbl>
## 1 (Intercept) 37.0      0.0168    2201.      0
## 2 age_years   0.000497 0.000506    0.982    0.326
## 3 feveryes    2.05     0.0163    126.      0
```



Summary

- When data is MAR, advanced imputation methods should be used
- Regression imputation uses a statistical model to impute missing values
- Multiple imputation creates multiple datasets with imputed values and uses them to calculate pooled estimates for analysis
- The **mice** function can be used to perform multiple imputation
- Both methods reduce bias and result in more accurate estimates

Managing missing data

Survey skip patterns

Heather Mattie, PhD

Surveys

Public health surveys are vital tools for understanding population health and informing public health policy

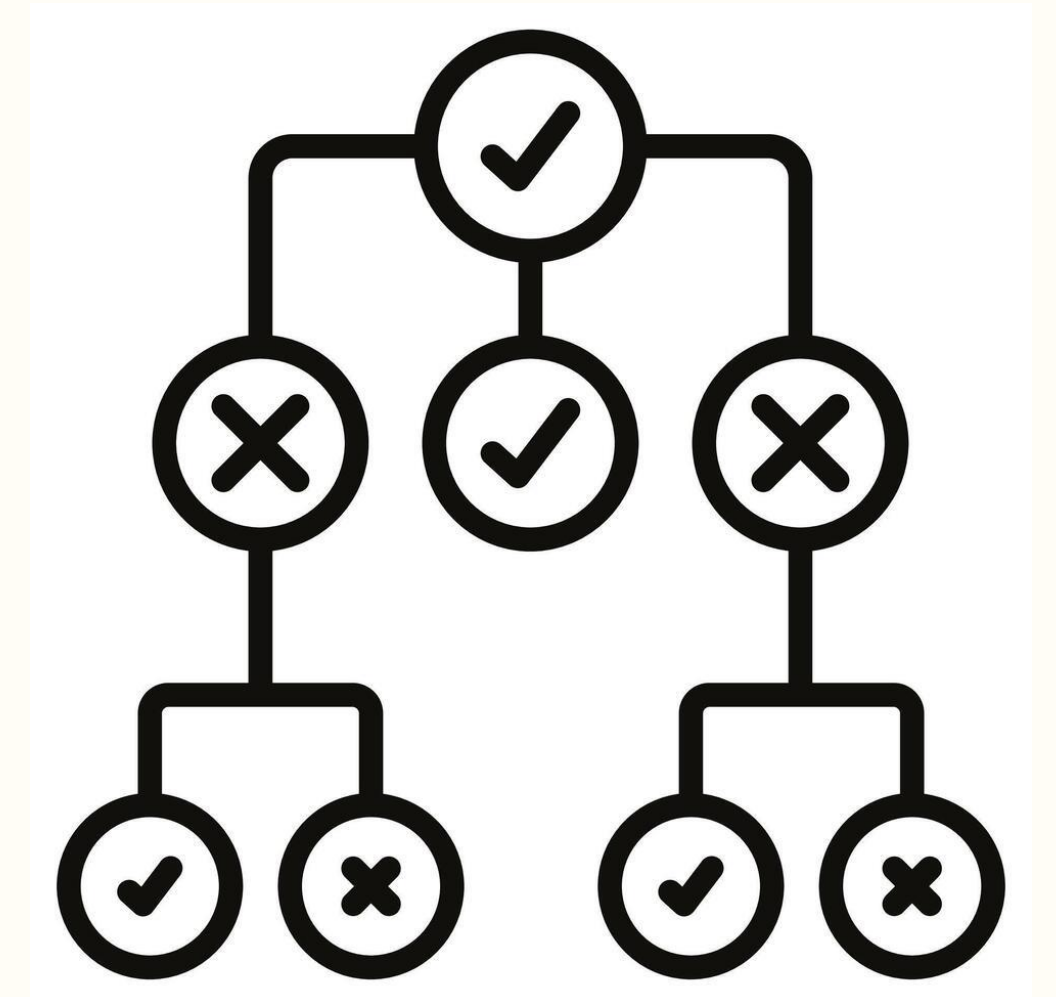
- Collect data on health status, behaviors, risk factors, etc.
- Allow researchers and policymakers to identify health disparities, track trends, and evaluate interventions
- Can be conducted at the national, state, or local level
- Can involve a combination of questionnaires, interviews, physical examinations, and lab tests



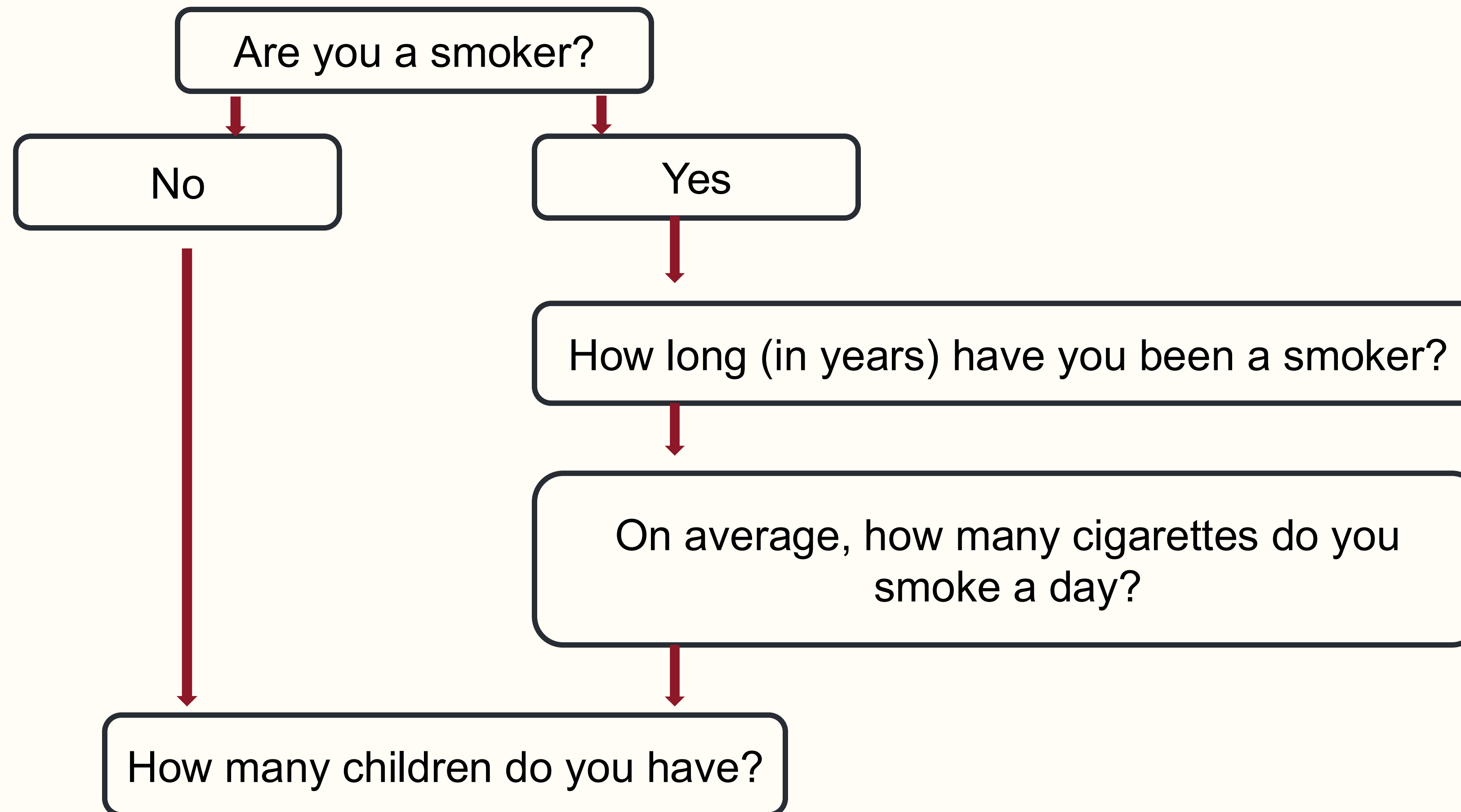
Skip pattern

A **skip pattern** in a survey is a feature that directs respondents to different questions or sections of a survey based on their answers

- Also called “skip logic” or “conditional branching”
- Ensures respondents only answer questions relevant to their specific situation
- Improves survey efficiency and engagement through streamlining
- Induces **missingness by design**



Skip pattern example



Skip pattern example

data

##	smoke	smoke_years	smoke_amt	children
## 1	Yes	24	20	0
## 2	Yes	7	15	1
## 3	No	NA	NA	4
## 4	No	NA	NA	2



Skip pattern example

data

##	smoke	smoke_years	smoke_amt	children
## 1	Yes	24	20	0
## 2	Yes	7	15	1
## 3	No	NA	NA	4
## 4	No	NA	NA	2

```
data <- data %>%
  mutate(smoke_years = ifelse(is.na(smoke_years), 0, smoke_years),
         smoke_amt = ifelse(is.na(smoke_amt), 0, smoke_amt))
```

data

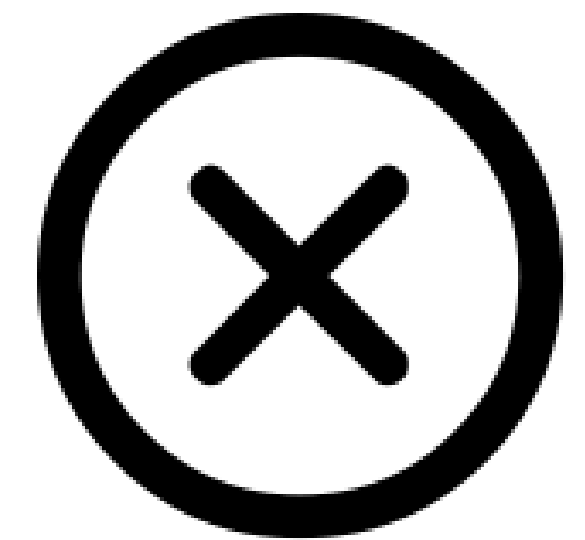
##	smoke	smoke_years	smoke_amt	children
## 1	Yes	24	20	0
## 2	Yes	7	15	1
## 3	No	0	0	4
## 4	No	0	0	2



Response bias

Response bias occurs when a respondent answers questions incorrectly or in a way that deviates from their true opinions or behaviors

- Many causes of response bias, we will focus on lack of understanding and accidental incorrect response
- Rather than removing these individuals from the dataset, can sometimes replace values that are known to be incorrect



Response bias example

Survey sent out to students at a university in the United States

- Included a question that asked if the respondent had received a federal grant
- This particular grant is only awarded to domestic students
- All international students *should* respond “No” to the federal grant question



Response bias example

data

##	international	grant
## 1	Yes	I don't know
## 2	Yes	Yes
## 3	Yes	No
## 4	No	Yes
## 5	No	No

Response bias example

data

##	international	grant
## 1	Yes	I don't know
## 2	Yes	Yes
## 3	Yes	No
## 4	No	Yes
## 5	No	No

```
data <- data %>%  
  mutate(grant = ifelse(international == "Yes" &  
                        grant != "No", "No", grant))  
data
```

##	international	grant
## 1	Yes	No
## 2	Yes	No
## 3	Yes	No
## 4	No	Yes
## 5	No	No

Summary

- Surveys are a common source of data in public health
- Survey skip patterns improve survey efficiency, but induce missingness by design
- Most skip pattern missing values can be imputed in a straightforward and logical way
- Response bias should be checked for before removing individuals from a dataset